Differentiable Simulation for Robotics

Literature Survey (2020-2026)

P(F) Shape Matching Debulking Project Reference

1. Differentiable Simulation Overview

Differentiable simulation means computing exact gradients of simulation output w.r.t. input parameters (tool trajectory, material properties, etc.) via automatic differentiation (autodiff). This enables gradient-based optimization of robot control, system identification, and design through physics.

Three Methods for Gradient Computation

Method	How It Works	Pros / Cons
Finite Difference	Perturb each param, re-run sim	Simple / O(N) sims per grad
Tape-based AD	Record ops, auto backward	Exact / Memory-heavy for long rollout
Analytical Gradient	Hand-derive adjoint equations	Fast & exact / Hard to implement
Implicit Diff (IFT)	Differentiate equilibrium eqn	Any horizon / Only for steady-state

2. Core Differentiable Simulation Frameworks

DiffTaichi: Differentiable Programming for Physical Simulation

ICLR 2

Taichi-based tape AD. 188x faster than TensorFlow. Supports elastic, cloth, rigid/soft coupling. Light-weight tape records program structure. Foundation for PlasticineLab.

Brax: A Differentiable Physics Engine for Large Scale Rigid Body Simulation

NeurIP

JAX-based massively parallel rigid body engine. 4 pipelines (MJX, Generalized, PBD, Spring). Includes PPO/SAC/ES. Differentiable through entire sim + learning loop.

NVIDIA Warp

2022-p

Python -> CUDA JIT. wp.Tape for reverse-mode AD. wp.svd3 differentiable. Supports PyTorch/JAX interop. Gradient checkpointing for memory. Production-grade performance. Used in Isaac Sim ecosystem.

Dojo: A Differentiable Physics Engine for Robotics

2022 |

Hard contact + friction via SOCP. Implicit differentiation (IFT). Julia with Python bindings. Applications: traj opt, policy learning, system ID.

Genesis: A Universal Physics Engine

2024 |

Universal engine: MPM, FEM, PBD, SPH, Rigid. Currently MPM differentiable. Photo-realistic ray-tracing. Positioning as

next-gen platform.

3. Differentiable Position-Based Dynamics (PBD)

DiffXPBD: Differentiable Position-Based Simulation of Compliant Constraint Dynamics

ACM S

XPBD with analytical gradient formulation. Handles 26M+ DOF meshes + collisions. No relaxation needed. Most directly relevant to our Shape Matching PBD framework. Shows PBD CAN be made differentiable with proper gradient derivation.

DiffPD: Differentiable Projective Dynamics

ACM T

Implicit time integration + Cholesky prefactorization. Exploits precomputed factorization to speed backward pass. Soft-body focused.

4. Differentiable Soft Body Simulation

PlasticineLab: A Soft-Body Manipulation Benchmark

ICLR 2

First soft-body benchmark with differentiable physics (DiffTaichi/MPM backend). Tasks: shape plasticine to target. 100-200 step rollout backprop works. Analytical gradients in soft-body benchmark.

DiffTactile: A Physics-based Differentiable Tactile Simulator

ICLR 2

FEM-based elastomer simulation. Multi-material (elastic, elastoplastic, cables). Penalty contact model. End-to-end differentiable.

SoftMAC: Differentiable Soft Body Simulation with Forecast-based Contact

2023

MPM soft bodies. Forecast-based contact reduces penetration. Two-way coupling: rigid + cloth + soft. Differentiable.

Fast and Reliable Gradients for Deformables Across Frictional Contact

2025

Analytical gradients validated vs FD. 33x speedup over finite difference. Handles soft AND stiff materials, small/large deformations.

5. Long-Rollout Gradient Stability (Core Challenge)

Backpropagating through 100+ simulation steps causes: (1) GPU memory explosion (tape stores all intermediate states), (2) gradient vanishing/exploding (500 matrix multiplications), (3) gradient discontinuity at contact events. This is the #1 open problem in differentiable simulation.

Mitigation Strategies

Strategy	Rollout Steps	Memory	Who Uses It
Truncated BPTT	10-50	Low	SAPO (2024), Brax
Checkpointing	100-500	Medium	Warp, DiffTaichi
Implicit Diff (IFT)	Any	Low	Dojo, contact-implicit MPC
RL + short grad	RL:inf, grad:10-50	Low	SAPO (2024) [BEST]

6. RL + Short-Horizon Differentiable Sim (Most Relevant)

SAPO: Stabilizing RL in Differentiable Multiphysics Simulation

2024 |

KEY PAPER for our approach. Short-horizon actor-critic: backprop only 10-50 steps, use RL for long-horizon. Handles rigid + articulated + deformables. Addresses exploding/vanishing gradients. Solutions: truncate stiff contacts, reweight/smooth gradients. Maximum entropy formulation (like SAC).

Accelerated Policy Learning with Parallel Differentiable Simulation

ICLR 2

Short-horizon actor-critic with truncated gradient backprop. Parallel environment simulation. Foundation for SAPO approach.

Learning Deployable Locomotion via Differentiable Simulation

ICRA 2

First successful sim-to-real transfer of legged locomotion learned entirely in differentiable simulator. Proves differentiable sim -> real robot works.

SAPO (2024) validates our approach: short-horizon autodiff (5-50 steps) + long-horizon RL (SAC). This is exactly what our P(F) framework enables.

7. Differentiable Surgical Simulation

VR-augmented Differentiable Simulations for Digital Twin

Nature

Digital twin for surgical planning. Trajectory generation in segmented medical images. VR overlay. Differentiable simulation for planning optimization.

Real-time Capable Learning-based Visual Tool Pose Correction via Differentiable Sim

2025

dVRK-compatible differentiable simulator. Supervised learning: tool segmentation -> pose corrections. Parallel computing for efficient gradients.

Warp-SurgicalSim (Our Reference Project)

Unpub

XPBD surgical sim with adhesion bonds. Claims "differentiable" but does NOT use wp.Tape or backprop through physics. Pure RL (SAC) training. Our P(F) framework has actual short-horizon differentiability -- a real advantage.

8. System Identification via Differentiable Sim

gradSim: Differentiable Simulation for System ID and Visuomotor Control

ICLR 2

Differentiable sim + rendering. Recovers physical properties from video. Parameters: friction, elasticity, material params. Backprop from pixels -> physics.

DPSI: Differentiable Physics-based System ID for Elastoplastic Materials

2024

Robots infer physics from manipulation + 3D point clouds. Directly relevant to our Task 2 (inverse tissue parameter estimation).

9. Our P(F) Framework Position in the Landscape

What We Have (Verified)

1. P(F) sigmoid damage model -- smooth, differentiable geometry sensing

2. Shape matching PBD in Warp with wp.svd3 -- differentiable SVD
3. Soft removal via sigmoid mask -- differentiable particle removal
4. Short-horizon gradient flow verified (5 steps, Phase 4)
5. Finite difference gradients correct direction (Phase 4)

Our Unique Contribution vs Existing Work

Existing Work	Our P(F) Framework
DiffXPBD: generic soft body	Surgical debulking with tissue sensing
PlasticineLab: shape deformation	Volume removal (debulking)
SAPO: generic RL + diff sim	P(F) geometry-aware reward
Warp-SurgicalSim: claims diff, not real	Actual short-horizon autodiff
Beihang SM: no differentiability	P(F) sigmoid is differentiable

Recommended Paper Framing

```
"Short-horizon differentiable surgical simulation with
P(F) geometry sensing, enabling analytic policy gradients
for sample-efficient RL training of debulking strategies."

Key claims:
1. P(F) provides differentiable, geometry-aware damage sensing
   (vs hard thresholds in Beihang/CRESSim)
2. Short-horizon autodiff through shape matching + P(F) sigmoid
   (verified, 5-step gradient flow)
3. Combined with SAC for long-horizon planning
   (following SAPO 2024 paradigm)
4. First differentiable surgical debulking simulation
   (Warp-SurgicalSim is not actually differentiable)
```

10. Key References

Paper	Venue	Link
DiffXPBD	ACM SIGGRAPH 2023	arxiv.org/abs/2301.01396
SAPO / Rewarped	2024	arxiv.org/abs/2412.12089
PlasticineLab	ICLR 2021	arxiv.org/abs/2104.03311
DiffTaichi	ICLR 2020	arxiv.org/abs/1910.00935
Brax	NeurIPS 2021	arxiv.org/abs/2106.13281
NVIDIA Warp	2022-present	nvidia.github.io/warp/
Dojo	2022	arxiv.org/abs/2203.00806
gradSim	ICLR 2021	arxiv.org/abs/2104.02646
DiffTactile	ICLR 2024	openreview.net/forum?id=eJHnSg783t
Genesis	2024	github.com/Genesis-Embodied-AI/Genesis
Fast Deformable Gradients	2025	arxiv.org/abs/2603.16478
Deployable Locomotion	ICRA 2024	arxiv.org/abs/2404.02887

Accelerated Policy Learning	ICLR 2022	arxiv.org/abs/2204.07137
Beihang SM GPU	CMPB 2024	ScienceDirect
VR Differentiable Digital Twin	Sci Reports 2025	nature.com/articles/s41598-025-99608-2